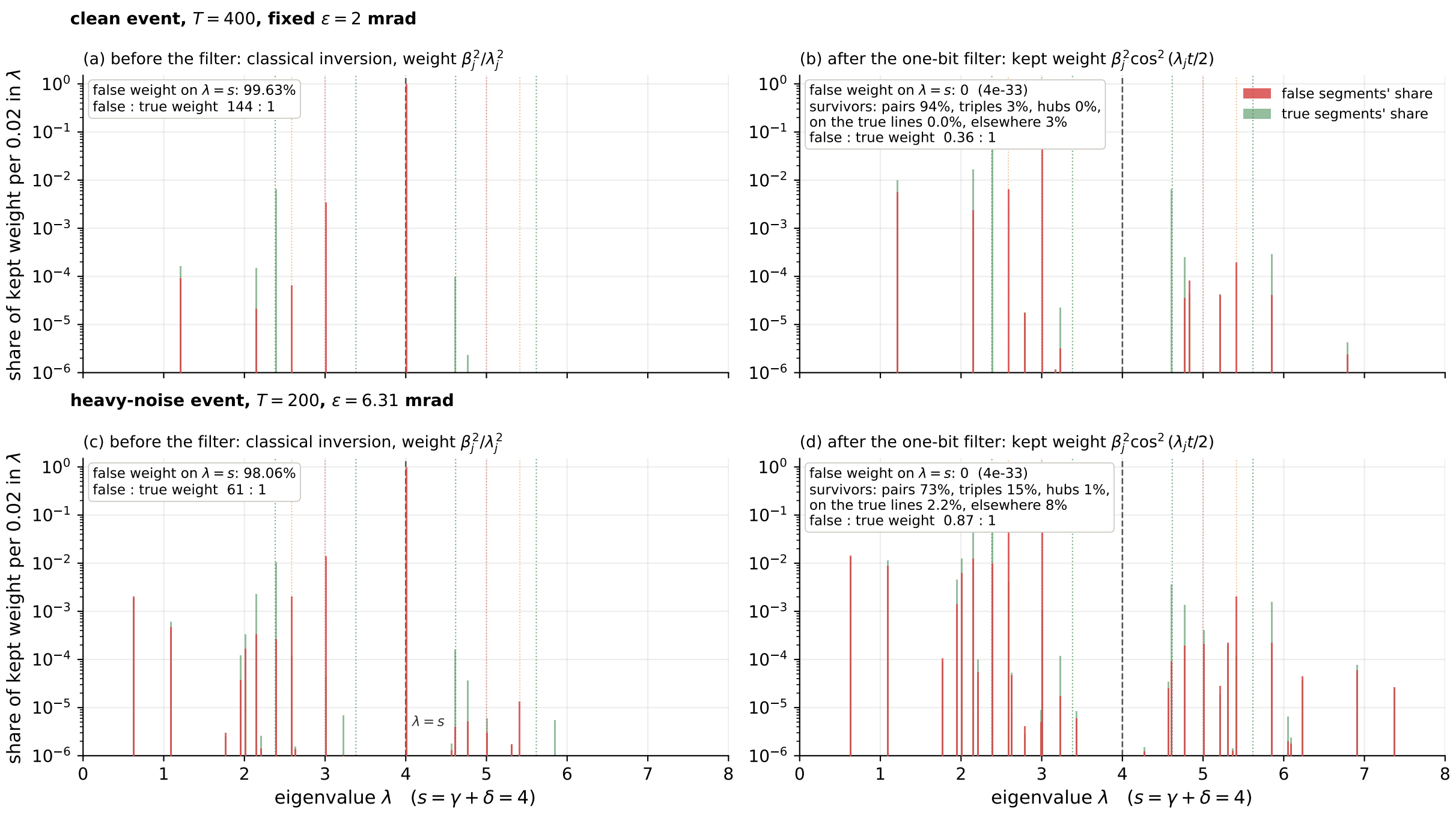

clean event, $T = 400$, fixed $\varepsilon = 2$ mrad
(a) before the filter: classical inversion, weight β_j^2/λ_j^2
false weight on $\lambda = s$: 99.63%
false : true weight 144 : 1
(b) after the one-bit filter: kept weight $\beta_j^2 \cos^2(\lambda_j t/2)$
false weight on $\lambda = s$: 0 (4e-33)
survivors: pairs 94%, triples 3%, hubs 0%,
on the true lines 0.0%, elsewhere 3%
false : true weight 0.36 : 1
false segments' share
true segments' share
heavy-noise event, $T = 200$, $\varepsilon = 6.31$ mrad
(c) before the filter: classical inversion, weight β_j^2/λ_j^2
false weight on $\lambda = s$: 98.06%
false : true weight 61 : 1
$\lambda = s$
(d) after the one-bit filter: kept weight $\beta_j^2 \cos^2(\lambda_j t/2)$
false weight on $\lambda = s$: 0 (4e-33)
survivors: pairs 73%, triples 15%, hubs 1%,
on the true lines 2.2%, elsewhere 8%
false : true weight 0.87 : 1
share of kept weight per 0.02 in λ
eigenvalue λ $(s = \gamma + \delta = 4)$